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Review article

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Towards an integrative neuroscience of metacognition

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Abstract

Metacognition enables organisms to self-evaluate their own cognitive processes. In humans, this capacity underpins adaptive learning, social coordination and the formation of self-beliefs about skill and ability. A core motif that underpins multiple forms of metacognition is the ability to form estimates of performance in a self-directed frame of reference, expressed as propositional confidence about one's own behaviour and mental processes. Convergent behavioural, computational and neural evidence obtained across rodents, non-human primates and humans indicates that these self-evaluations arise from structured transformations of uncertainty. Dynamic evidence accumulation processes provide candidate building blocks for local confidence, whereas multimodal prefrontal brain areas integrate these signals to support abstract, global self-beliefs. This architecture supports metacognitive control at multiple timescales, from rapid error correction to strategic task arbitration and social communication. Dysfunctional metacognition can lead to disorders of reality monitoring and insight, in which erroneous mental representations are endorsed with aberrant confidence. By integrating computational, cross-species and clinical perspectives, metacognitive neuroscience offers a bridge from neural circuit mechanisms to subjective experience.

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Introduction

Metacognition is a broad, umbrella term for the processes that allow us to reflect on and self-evaluate mental and physical function and use these insights to control behaviour[1]. When lining up to take a penalty kick in football, you might estimate your chances of precisely placing the ball in the top-right hand corner of the goal. If you realize that you are not up to the task, you could hand over the task to a more confident teammate. Or, when driving down a foggy road at night you might deliberately decide to slow down, because you realize that the visual information that you have available is not sufficient to efficiently stop at higher speeds. These examples illustrate the intimate connections between metacognitive self-evaluations and adaptive control, guiding how we seek information and advice, how we regulate our learning and collaborate with others, and how or whether we offload to external aids.

Metacognition research has its origins in cognitive psychology, particularly within educational contexts in which researchers have been interested in how children develop beliefs about their own knowledge[2]. Historically, metacognition was considered a uniquely human faculty. More recently, it has become possible to develop task-based measures and computational metrics of metacognition that can be deployed across both humans and non-human animals[3,4] — metrics that are now proving useful for probing metacognition in artificial intelligence systems (Box 1). Previous review articles have focused on psychological[5–7] and computational[8,9] aspects of metacognition; here, I focus on its neural basis, with the aim of integrating findings from systems and cognitive neuroscience.

In this Review Article, the term metacognition is used to refer primarily to metacognitive monitoring, as this aspect has received the most attention from neuroscientists. Metacognitive monitoring refers to the ongoing assessment of the reliability or quality of one's own mental processes — such as perception, memory, reasoning or decision-making — and typically involves the generation of second-order representations of confidence about a first-order cognitive process. Confidence formation is therefore a core variable of interest for metacognition research. Metacognitive monitoring can be harnessed for adaptive metacognitive control (Fig. 1a), in which metacognitive estimates are used either for regulating behaviour and/or for communicating metacognitive assessments to others[10]. I turn to these functional roles of metacognition later in the Review. From a functional standpoint, introspection and metacognition are similar psychological concepts. However, because the term introspection tends to presume a connection with consciousness, I use the more neutral term metacognition here, and instead consider in Box 2 how metacognitive mechanisms may themselves be important for developing functional explanations of consciousness.

I begin by articulating the quantitative approaches that can be used to define and measure metacognition in the laboratory (Fig. 1b). I then aim to synthesize results obtained in rodents, monkeys and humans that illuminate the neural basis of canonical metacognitive judgements and showcase how a network of brain regions, centred on hubs in the prefrontal cortex (PFC), build up beliefs about self-performance via interactions with domain-specific and modality-specific areas. I ask how metacognitive processes that are expressed on a local, moment-to-moment timescale shape the formation of global self-beliefs about skills and abilities over longer timescales (Fig. 1c). Finally, I ask how distortions in metacognitive processes may underpin a variety of mental health symptoms in which there is a disconnect between subjective reality and objective performance.

Measures of metacognition

Multiple forms of metacognition are underpinned by the ability to form estimates of uncertainty in a self-directed frame of reference, with these estimates expressed as propositional confidence about our own behaviour and mental processes[7]. This capacity allows us to estimate feelings of surety in our own (hypothetical) decisions or actions, resulting in covert propositions such as "I think I will remember this word." Propositional confidence can be distinguished computationally from other forms of world-directed or distributional uncertainty[11,12]. The self-directed nature of metacognition makes it challenging to design tasks that truly tap into metacognitive processes, as not all behaviour that seems metacognitive is metacognitive in nature[13]. For instance, selectively avoiding difficult choices may be driven by properties of the stimulus or situation, rather than representing to oneself that the decision is difficult.

Tasks designed to measure metacognition require two components: a first-order behaviour or mental state to be monitored and a way of eliciting a second-order evaluation of the first-order behaviour or mental state (Fig. 1b). In humans, these judgements are usually explicit: subjects are instructed to press a button or select a point on a scale to indicate their confidence. In other cases, such as confidence forced-choice paradigms, they are required to select the decision that they feel most confident about from a pair of decisions[14]. In animal metacognition research, confidence estimates are elicited via so-called implicit measures of confidence, described further below. When we are comparing the literature on metacognition across species, it is important to keep in mind that behavioural and neural markers of metacognition in animals may be tapping into implicit forms of metacognition: whether there is a cross-species analogue of explicit metacognition remains unclear[15].

A popular approach to assess metacognition in non-human animals harnesses opt-out paradigms, in which the animal is initially trained to produce a first-order behaviour (for instance, a perceptual discrimination judgement) and then trained to select an additional opt-out response when uncertain about its primary decision[4]. Critics of opt-out paradigms highlight ambiguity as to whether they are being solved in a world-directed or self-directed manner, because an effective opt-out decision can be made by tracking stimulus strength, rather than performance conditional on a chosen action[15]. Indeed, on opt-out trials, it may be the case that a primary decision was never formed. However, if the opportunity to opt-out is offered randomly and intermittently and is unknown to the decision-maker when viewing the stimulus, provisional formation of a primary decision is more likely[16].

An alternative approach has used variants on post-(or peri-)decision wagering — in which subjects make incentivised wagers on their performance, elicited as a second-stage behaviour (such as a button press) after each trial of the first-order task. The typical setup of a post-decision wagering experiment involves training the animal or human to understand that a high confidence response is risky, as it will only be highly rewarded when correct, whereas a low confidence response is guaranteed to receive a smaller reward irrespective of response correctness. This creates incentives to engage in metacognitive monitoring to wager effectively. In a variant of the post-decision wagering paradigm developed for use with rodents[3] and human infants[17], subjects are trained to wait for an uncertain reward that is more likely to be delivered when they are correct, and withheld when incorrect. At any point in this waiting period, they can give up to start a new trial. The optimal strategy here is to wait longer when confidence in being correct (and therefore the subjective probability of a reward

Box 1 | From biological to artificial metacognition

Given the importance of metacognition for monitoring and evaluating knowledge and performance it has become increasingly pressing to understand the metacognitive capacities of artificial intelligence (AI) systems[187]. The incorporation of effective metacognition into generative AI systems such as large language models has been historically difficult to achieve, with notable disconnections between confidence and accuracy in safety-critical domains such as medical decision-making[188]. This can lead to a 'calibration gap', whereby humans assume artificial systems are more accurate than they really are[189]. There is, however, growing evidence that artificial neural networks can form and update calibrated confidence estimates[87], estimate their own knowledge states[190], identify their own behavioural propensities[191,192] and potentially have access to their own internal workings[193,194] — albeit with profiles that may diverge substantially from both normative standards and human judgement[195]. Artificial neural networks also provide a useful testbed for exploring the functional profile of different metacognitive architectures. For instance, one study found that idiosyncratic biases in confidence emerge within a convolutional neural network trained to discriminate handwritten digits — indicating that such biases may not be unique limitations of human metacognition but rather a core feature of how high-dimensional evidence spaces are mapped to confidence[87].

The mechanisms supporting such 'introspection' in artificial systems are likely to be very different from architectures supporting human metacognition[196] and will depend on the tasks the systems have been optimized to solve. If a main function of introspective systems is to allow performance prediction and error monitoring, then a low-dimensional metacognitive signal may suffice. But if there is a functional need to communicate the internal states of the system to others, then a more granular introspective system may be required. In either case, the development of such metacognitive capacities in AI systems will likely be important in managing both self and social regulation in human–AI interactions[187]. In turn, as the first-order capabilities of AI become more sophisticated, the demands placed on human metacognition will increase, as people will be required to know when and how to efficiently offload cognitive work to AI and monitor its output[197].

being available) increases[18]. Figure 1b illustrates the various approaches that have been used to measure metacognition.

Once first-order and second-order behaviours have been elicited over multiple trials or episodes of a task, these data can be analysed to quantify a metacognitive profile. The field has made use of two broad classes of summary statistic[19]. The first is metacognitive bias — the level of a metacognitive assessment (for instance, high or low confidence) relative to performance (this is sometimes referred to as calibration or simply confidence level). The second is referred to as metacognitive sensitivity — the extent to which metacognitive judgements correlate with behavioural performance from trial to trial. Estimates of metacognitive sensitivity are easily confounded with changes in first-order performance or metacognitive bias[19]. When sensitivity is corrected for these confounds, it is sometimes referred to as metacognitive efficiency. A popular approach to distinguishing bias and sensitivity in metacognitive judgements is to apply the meta-d' model[20], which grew out of established signal detection theory (SDT) approaches within cognitive psychology. SDT provides a simple framework for measuring how humans and animals make decisions under uncertainty across a range of tasks. A classical SDT analysis returns a sensitivity parameter, d', which reflects a participant's performance on the first-order perception, memory or decision task. Meta-d' reflects the first-order sensitivity parameter that best explains an agent's confidence ratings under a lossless SDT model. Meta-d' is a descriptive statistic and admits of various computational architectures underpinning better or worse metacognition.

Neural basis of metacognition

Neural correlates of metacognition: a brief history

A first wave of studies starting in the 1980s identified putative neural correlates of metacognition through observational studies relating metacognitive sensitivity to features of brain structure and function. Shimamura and Squire[21] found that people with Korsakoff's syndrome (a chronic neurological disorder associated with alcohol misuse) had selective impairment in the accuracy of prospective metacognitive judgements of memory compared to an amnesic control group, despite recognition memory performance being similar. This research identified the human PFC as a key neural substrate for metacognition[22] and underpinned subsequent exploration of the neural correlates of metacognitive sensitivity using performance-corrected tools and metrics. Early investigations of individual differences in metacognitive sensitivity also added weight to the proposal that the PFC supports metacognitive capacity, identifying links with its structure[23–25], function[26,27] and connectivity[28–30]. This work was complemented by studies of the lifespan trajectory of metacognition, showing that metacognitive sensitivity matures through adolescence[25,31], plateaux in adulthood[32] and can remain robust into older age despite other aspects of task performance deteriorating[33,34].

A second wave of studies beginning in the 2000s used functional brain imaging to chart the neural correlates of metacognitive judgements. A first approach, in humans, compared the requirement for a metacognitive judgement to a control condition, highlighting selective increases in activation within the medial frontal cortex (specifically, the dorsal anterior cingulate cortex) and lateral frontopolar cortex (FPC)[26,35,36]. A second approach sought to relate changes in neural activity to changes in the level of a metacognitive judgement — for instance, correlating the degree of confidence with increases or decreases in the BOLD signal or neural firing rate. Here again, studies identified neural correlates of confidence in the frontal and parietal cortex of monkeys[16,37,38], humans[26,39,40] and rodents[41]. In rodents, lesions to the orbitofrontal cortex (OFC) and ACC have been shown to reduce metacognitive sensitivity, while leaving first-order performance intact[42,43]. In both humans and non-human primates, lesions or inactivation of the FPC have been linked to selective metacognitive deficits[38,44–47] (Fig. 2), consistent with its position at the apex of a neurocognitive hierarchy[48].

The earliest work in this second wave predominantly focused on metacognitive judgements about percepts and memories that have objectively correct or incorrect answers — truth-conditions that can be

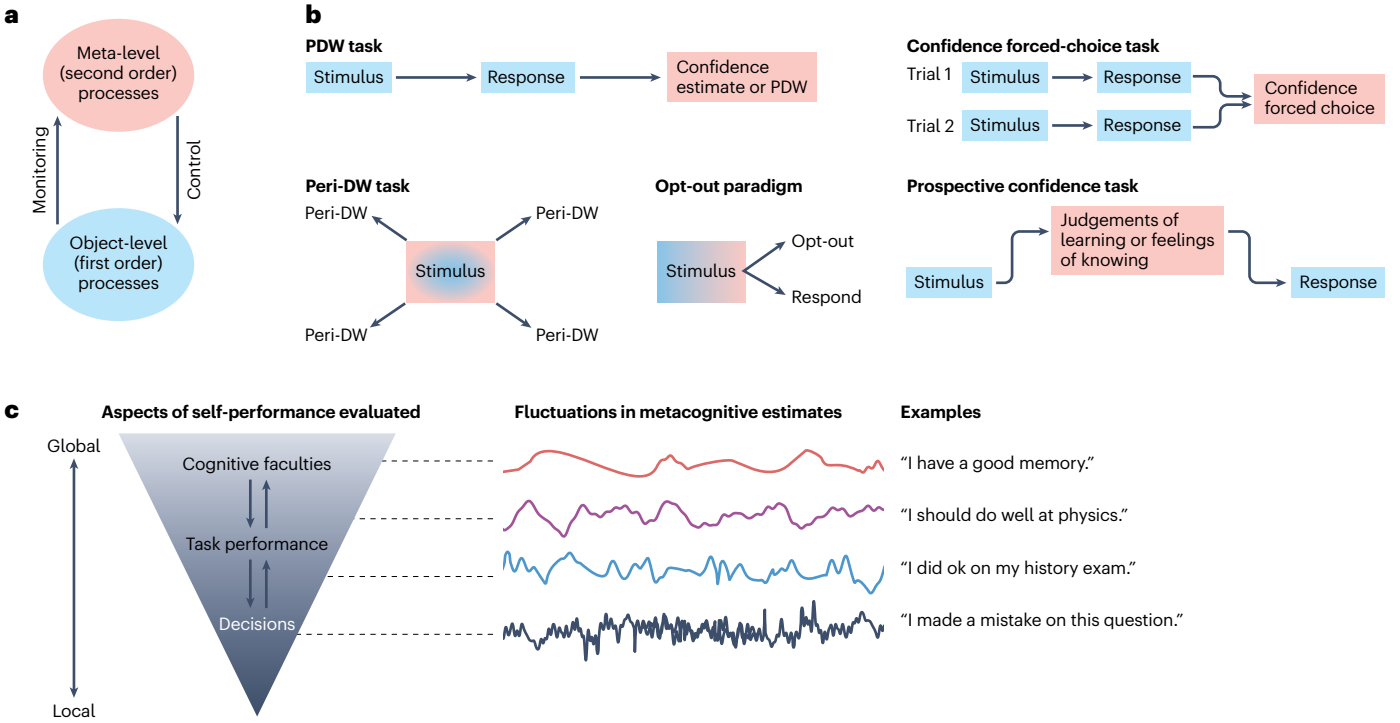


Fig. 1 | Tasks and frameworks for characterizing metacognition. a, The classic Nelson and Narens model of metacognition, incorporating monitoring and control[10]. Meta-level (second-order) processes are proposed to monitor and control first-order (object-level) mental processes. **b**, A range of different paradigms have been developed for the measurement of metacognition across species. In retrospective confidence/post-decision wagering (PDW) tasks a confidence estimate or post-decision wager is elicited following a first-order task response. In a confidence forced-choice procedure the participant is asked to choose which of two trials they performed better on. In a peri-decision wagering (peri-DW) task the first-order response and the confidence estimate are elicited simultaneously. In an opt-out paradigm, participants can either choose to respond or opt-out conditional on their assessment of the difficulty of the trial. Other tasks measure confidence about future decisions or actions (prospective confidence), using measures that include judgements of learning or feelings of knowing. The red and blue shading roughly corresponds to the stages of the task behaviour that map onto first-order and second-order components of the model in panel **a**. **c**, A hierarchical framework for metacognition posits that reciprocal interactions between local and global self-evaluations unfold over different timescales[101]. Estimates of self-performance shown on the left span from local, moment-to-moment evaluations of individual decisions and actions up to more global, long-run evaluations of cognitive faculties. The traces in the centre illustrate the fluctuations in metacognitive estimates at different levels over time, with reciprocal and interactive relationships between levels. Right: examples of the ways in which these estimates might be expressed are shown.

verified by the experimenter. However, in daily life we often self-evaluate decisions without an obvious correct answer — so-called value-based or subjective decisions. By asking for explicit judgements of confidence in subjective decisions (such as which of two snack items people preferred to eat), it was possible to show that people could recognize suboptimal subjective choices[29]. Notably, confidence in value-based decision-making exhibited similar neural correlates to those found in studies of perceptual and mnemonic metacognition: whereas the ventromedial PFC carried multiplexed confidence and value signals, the FPC specifically tracked both confidence and individual differences in metacognitive sensitivity[29,49].

When interpreting studies of the neural correlates of confidence, performance confounds must be taken into account. Fluctuations in confidence are often associated with changes in both stimulus strength and choice accuracy, rendering it ambiguous as to whether the observed neural response is covarying with factors affecting first-order performance, metacognition, or a mixture of the two. The advantage of computational frameworks is that they make explicit the latent processes involved, allowing us to discover the building blocks of a canonical metacognitive judgement and dissociate them from factors driving first-order task performance[50]. Below, I consider how findings on the neural basis of metacognition relate to hypotheses derived from different computational accounts.

Neurocomputational approaches to metacognition

The simplest models of metacognition, sometimes referred to as direct access or first-order models, assume that the subject has perfect access to the evidence or process underpinning their first-order behaviour. According to these accounts, there should be minimal dissociations between performance and metacognition, and the evidence used for both is computed in the same (world-directed) frame of reference. Indeed, such accounts are not really metacognitive, in that apparently metacognitive behaviour is being explained by a first-order (non-metacognitive) process. However, first-order accounts struggle to accommodate cases in which the psychological or neural characteristics of metacognition diverge from those supporting first-order performance[51]. In such cases, additional computations supporting metacognition need to be elucidated.

Two broad cross-cutting distinctions help organize these more advanced computational approaches to confidence and metacognition. First, we can divide models into static and dynamic, based on whether or not they seek to model the latencies of task performance and metacognitive judgements. Second, in both static and dynamic models we can identify serial, parallel and hierarchical architectures relating performance to metacognition[8,51].

Static models of metacognition. Static models of metacognition seek to characterize quantitative relationships between performance and self-evaluation while remaining agnostic as to the dynamics of these relationships over time (either within a trial, or over multiple trials). Static models are useful because they generate statistical signatures of confidence formation — data patterns that are indicative of metacognition in a way that cannot be easily explained by other underlying decision processes.

A useful statistical signature of confidence that can be straightforwardly derived from static models is the so-called 'folded-X' pattern[18]. When plotted against objective measures of stimulus strength, propositional confidence should increase with evidence strength for correct trials, and decrease with evidence strength for error trials (Fig. 3a). A folded-X pattern has been observed in human and animal experiments designed to assess confidence[18] and used as a non-verbal indicator that physiological or neural signals are confidence-related[52]. For example, in a seminal study, the activity of neurons in the rodent OFC was shown to exhibit a folded-X pattern in an odour discrimination task[41] (Fig. 3b). This activity pattern predicted post-decision wagering behaviours and generalized across both auditory and olfactory decisions[53]. However, recent findings suggest that whether or not a folded-X pattern is observed is context-dependent: manipulations of evidence strength that allow the observer to infer task difficulty, such as stimulus reliability or visibility, can lead to increases in confidence for both correct and error trials[54,55].

Other static models have sought to isolate the noise or imprecision in metacognitive judgements from aspects of first-order task performance. In the 'confidence as a noisy decision-reliability estimate' or CASANDRE model, metacognition is proposed to be limited by the extent to which individuals have meta-uncertainty about the world-directed uncertainties affecting their task performance[56]. Meta-uncertainty can be estimated from fits to psychometric functions that reveal how stimulus strength differentially influences performance and confidence. Fitted meta-uncertainty is partly domain-general, showing cross-task correlations that are reduced as the distance between tasks in task space increases[56,57], and has been estimated in both humans and monkeys[58]. CASANDRE is a hierarchical, static model, in that a ceiling on metacognitive performance is provided by the degree of meta-uncertainty. By contrast, parallel architectures such as the 'confidence-noise confidence-boost'

Box 2 | Perceptual metacognition and consciousness

Consciousness refers to subjective experience. Although metacognition and consciousness are distinct concepts, there is growing interest in the role of metacognitive processes in computational explanations of human consciousness. This approach has its origins in higher-order theories of consciousness, in which a central aspect of becoming conscious of a mental state is the creation of a metacognitive representation of that mental state[198]. Metacognitive sensitivity has been shown to be 'consciousness selective', in that people are typically better able to monitor their performance when they are also conscious of the relevant information[185]. One proposal that accounts for this linkage holds that percepts become conscious when they are deemed to be strong or reliable enough to impinge upon cognition[117,199].

A core metacognitive operation that is relevant to this argument is known as 'reality monitoring' — determining which aspects of internal neural activity track an individual's current environment and which are engaged in internal aspects of planning, simulation and so forth. Studies have shown that mnemonic reality monitoring — figuring out whether something had previously been experienced or just imagined — recruits similar regions of the frontopolar cortex as explicit metacognition[200]. Recent work has developed paradigms for studying perceptual reality monitoring by asking subjects to imagine low-level visual stimuli embedded in dynamic noise, and then occasionally 'fading in' a real stimulus. When imagination is congruent with external stimulation and when it is particularly vivid, subjects are more likely to attribute their experience to reality — suggesting that a central driver of reality monitoring is the strength and precision of ongoing perceptual processing[201]. Using functional magnetic resonance imaging, it was found that anterior insula activity both correlated with reality judgements and tracked fluctuations in the strength of mid-level visual cortical activity, furnishing a potential neural substrate for the higher-order monitoring of sensory strength[202]. There is natural overlap here with work on neural 'read-outs' of sensory uncertainty.

The metacognitive computation involved in reality monitoring is likely implicit — we are not typically explicitly aware that we are engaged in this process, except in special circumstances (for instance, when trying to figure out whether a seemingly illusory experience is real). Accordingly, people have limited insight into the trial-to-trial confusions generated in perceptual reality monitoring experiments[203]. Instead, gaining insight into one's hallucinatory experiences may depend on additional layers of explicit metacognition[204].

Within this framework, aphantasia — a lack of conscious imagery, despite intact performance on a range of tasks — may reflect a change in metacognition. The idea is that people can still internally activate the relevant perceptual representations to solve a task, but higher-order metacognitive representations (correctly) discard them as internally generated, and they are not consciously experienced[154]. Indeed, people with aphantasia have been suggested to exhibit a dysconnection syndrome, in which regions of the brain implicated in perceptual metacognition (such as the anterior prefrontal cortex) are decoupled from sensory areas[205], potentially underpinning this change in conscious experience. It is challenging to develop animal models of distortions in reality monitoring. However, one promising approach combines measures of metacognitive confidence with perceptual tasks, in order to identify confident but false percepts[206].

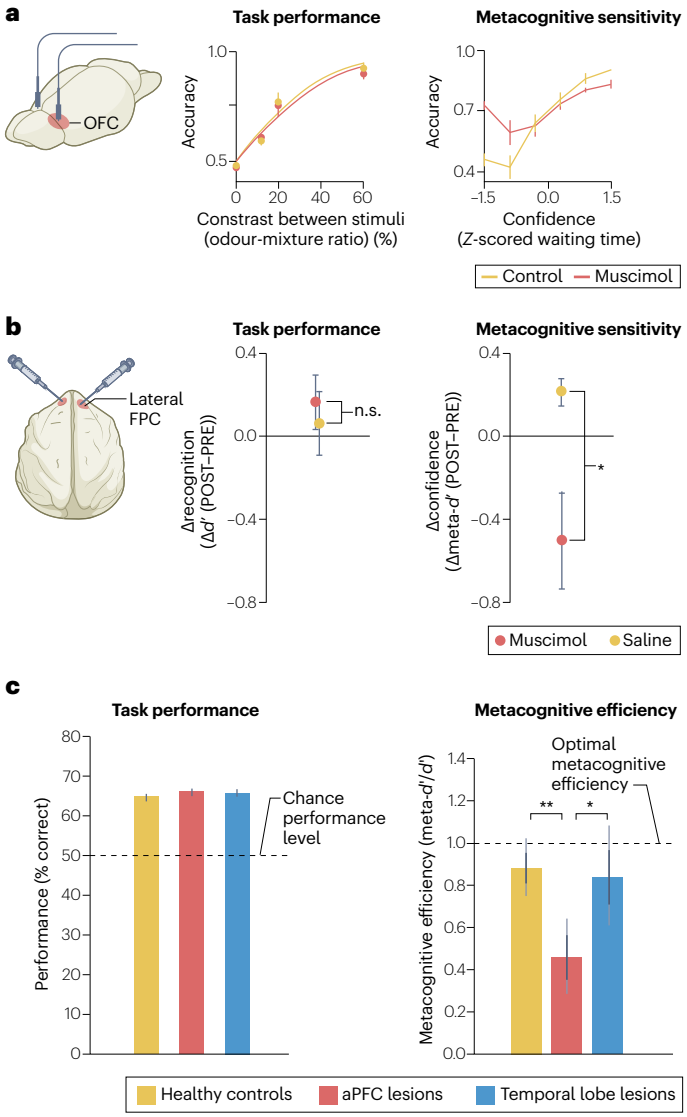


Fig. 2 | Convergent evidence for a role of prefrontal cortex in metacognition across species. a, In rodents completing a post-decision wagering task in which confidence is indicated by the time the animals are willing to wait for reward delivery, reversible inactivation of the orbitofrontal cortex (OFC) via infusion of the GABA-A receptor agonist muscimol leads to a deficit in metacognitive sensitivity (the relationship between confidence and accuracy) (right) without changing task performance (left)[42]. **b**, In monkeys engaged in a post-decision wagering task, reversible inactivation of the lateral frontopolar cortex (FPC) via injection of muscimol leads to a deficit in metacognitive sensitivity (right) but not recognition memory task performance (left)[38]. PRE and POST refer to before and after the injection, respectively; meta-*d*′ is a measure of metacognitive sensitivity extracted from a signal detection theory model. **P* < 0.05; n.s., not significant. **c**, In humans, lesions to the anterior prefrontal cortex (aPFC) are associated with deficits in metacognitive efficiency for perceptual judgements (right) despite intact perceptual task performance (left)[44]. ***P* < 0.01; **P* < 0.05. Panel **a** adapted with permission from ref. 42, Elsevier. Panel **b** adapted with permission from ref. 38, Elsevier. Panel **c** adapted with permission from ref. 44, CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/).

model allow for independent sources of evidence to contribute to a confidence judgement, such that metacognition may be 'boosted' relative to performance[14,51]. Combining these static models with neural measures offers powerful opportunities for future work on the neural basis of metacognition[58].

Dynamic models of metacognition. A limitation of static models of metacognition is that they do not make predictions about the timescales of performance and self-evaluation, restricting the extent to which they can be related to the dynamics of brain activity. Just as static models have often been grounded in SDT, most dynamic models of metacognition have built on evidence accumulation (EA) models. EA models propose that the decision-maker accumulates noisy evidence in favour of different choice options until a threshold is reached and a decision is made[59]. EA models were initially developed to account for first-order perceptual and mnemonic decision-making but can be extended to accommodate confidence by assuming that metacognitive evaluation rests on extracting the degree of evidence supporting the first-order decision[60]. From this shared starting point, models diverge as to their core assumptions about the mechanisms for generating confidence estimates (Fig. 3c).

Mirroring the divide between hierarchical and parallel static models, distinct model variants assume either serial or parallel EA for confidence. Serial models propose that, after a decision, additional post-decisional accumulation continues, capturing time-dependent fluctuations in subsequent confidence level[61–63]. By contrast, parallel models allow confidence to be continuously 'read out' from the evidence available during the pre-decision phase[16]. There is evidence for behavioural and neural correlates of both of these processes (Fig. 3d).

In support of parallel models, the dynamics of EA in the pre-decision phase have been shown to predict confidence levels. In monkeys, the rate of EA in lateral intraparietal cortex firing rates has been linked to confidence measured using both opt-out choices[16] and peri-decision wagering[64]. In humans, research using electro-encephalography (EEG) has found that the centroparietal positivity, an event-related potential component linked to EA in perceptual decision-making, predicts confidence level measured via explicit ratings[65], including those measured using psychophysical techniques to dissociate confidence from performance, reaction time and stimulus strength[66].

There is also evidence for a parallel encoding of confidence. In a recent study in monkeys, peri-decision wagering was used to simultaneously measure choices and confidence in a motion discrimination task[64]. The authors found that the monkeys' behaviour was well-fit by an EA model in which choice and confidence are evaluated concurrently, but not by a serial choice-then-confidence model. Single-trial decoding of neural activity in the lateral intraparietal cortex revealed that choice and confidence-predictive signals emerged simultaneously at the start of the trial, consistent with the parallel evolution of task performance and metacognition, and with previous research proposing that decisions and confidence estimates are encoded in parallel cortico-thalamic loops[67]. In an experiment using human magnetoencephalography (MEG), a similar decoding profile was observed, with both choice and confidence signals emerging in parallel at stimulus onset, and occupying different sensor subspaces (implying distinct neuronal populations) across the frontal and parietal cortex[68]. These parallel signatures of EA for choice and confidence may explain the capacity to dynamically develop metacognitive estimates 'online' for use in the rapid correction of errors[69–71] and the control of decision policies[72,73].

Finally, there is evidence for post-decisional processing contributing to confidence formation[74], consistent with a serial architecture.

Much of this work has been influenced by early discoveries about the neural basis of error monitoring. A canonical finding is that posterior medial frontal cortex (pMFC) neurons activate when errors are made in both macaques[75] and humans[76], generating a classical error-related negativity at the scalp surface[74,77]. A later event-related potential component, the Pe, has been linked to explicit metacognitive reports of error[78,79] and is thought to be generated by the insula cortex[80]. More recently, work in humans using EEG has revealed that decision errors trigger a second phase of EA (linked to a similar centroparietal positivity potential associated with pre-decisional EA) that can explain features of confidence, error monitoring and changes of mind[71,74,81,82]. Similarly, using functional magnetic resonance imaging (fMRI), regions commonly implicated in error monitoring such as the pMFC have been shown to harbour graded signatures of evidence for or against a previous choice[83].

Some of the debate over parallel and serial architectures for confidence reports may be due to the timing of the metacognitive judgement in different tasks. When metacognitive judgements are collected retrospectively, the importance of a post-decisional phase of EA is emphasized[37]. By contrast, when metacognitive judgements are elicited concurrently with a choice (as in peri-decision wagering paradigms), emphasis is placed on parallel signatures of confidence. As has previously been noted, there is no reason that post-decisional contributions to confidence judgements should be mutually exclusive of a provisional confidence estimate emerging during the decision process[8]. In practice, in naturalistic settings, both pre- and post-decisional dynamics are likely to contribute to the continuous evolution of the metacognitive state of the system.

Sources of information for metacognitive computation. What information informs these neural signatures of propositional confidence? In the previous section we explored static and dynamic models that characterize how latent decision variables (such as EA or meta-uncertainty) influence confidence formation. But in isolation, these models are largely silent on the kinds of sensory and mnemonic representations that inform these computations.

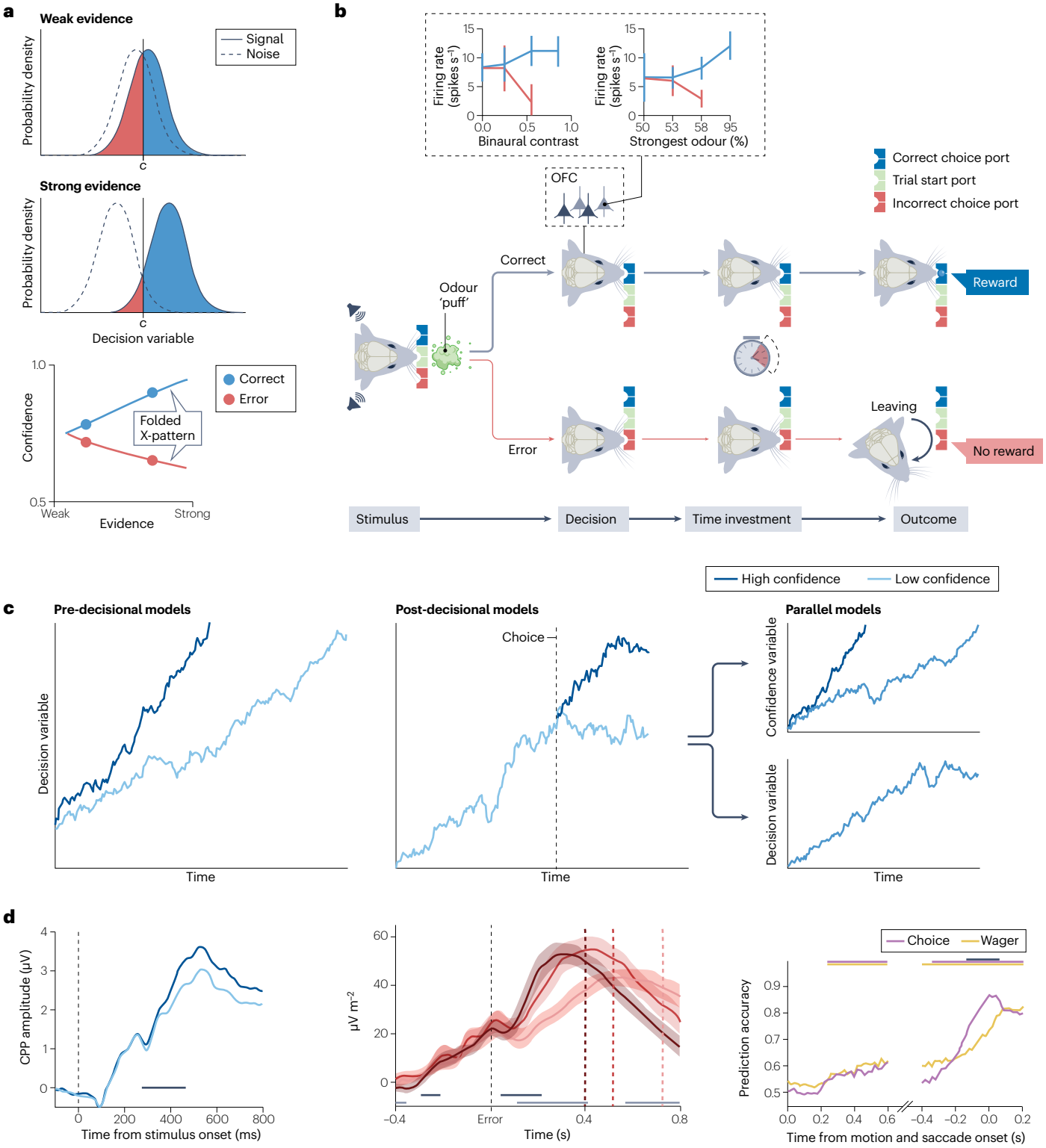
One prominent proposal is that, within a given task, world-directed uncertainty needs to be extracted by brain areas downstream of sensory regions to enable metacognitive behaviours[12,56]. World-directed uncertainty refers to external quantities, such as sensory uncertainty about line orientation, or cognitive uncertainty about whether Manchester United will win the English Premier League. These uncertainties are not themselves metacognitive – they are about aspects of the world, rather than the self – but they provide crucial inputs to metacognitive judgements (Fig. 4a). For instance, when making a judgement about line orientation, it is proposed that a summary of the implicit or 'distributional' uncertainty can be extracted from the activity of neural populations in visual cortex tracking line orientation[84]. Similarly, in an auditory task, the relevant uncertainties are suggested to be implicitly encoded by populations of auditory cortex neurons[85].

Recent work has sought to understand how the neural encoding of world-directed sensory, mnemonic and motor uncertainty is connected to the downstream neural mechanisms underlying self-evaluative judgement. To this end, a machine learning approach has been used to decode trial-by-trial uncertainty in the representation of the orientations of a visual grating from fMRI data collected from the human visual cortex. Reported confidence was negatively correlated with the decoder's readout of uncertainty, even for identical stimuli[86]. In turn, the decoder's readout of sensory uncertainty was correlated with fMRI signals in the PFC, suggesting that domain-specific uncertainty estimates in sensory regions inform downstream representations of propositional confidence in prefrontal regions. In another recent study, monkeys were trained on an orientation discrimination task and reported their confidence using a peri-decision wagering procedure[58]. Although both linear and nonlinear decoders applied to neuronal activity in the primary visual cortex predicted first-order choices, nonlinear decoders consistently outperformed linear decoders in predicting the monkey's confidence estimates, suggesting that confidence arises from an additional transformation of the sensory signals that inform perceptual decisions[87]. Similarly, another fMRI study showed that a proxy for sensory uncertainty (motion coherence, which was dissociated from choice difficulty) was related to activity in the extrastriate visual and parietal cortex, whereas activity related to propositional confidence was observed within the perigenual anterior cingulate cortex (pgACC)[88]. These results are consistent with the idea that world-directed visual uncertainty (tracked in visual and parietal cortex) is converted to a self-directed format in downstream PFC.

It remains unclear whether similar mechanisms apply beyond sensory and motor representations, contributing, for example, to confidence in being able to remember something. A recent study in macaque monkeys recorded from neural populations in the lateral PFC during a spatial working memory task. Population activity in this region during the delay period in which the animals were required to remember the locations of the stimuli jointly encoded both the remembered locations and their associated uncertainty, with a decoded measure of working memory strength predicting both recall accuracy and opt-out decisions on a trial-by-trial basis. Notably, a distinct metacognitive signal – reflecting the decision to opt-out – emerged approximately 200 ms after the working memory representation in a different subspace of the same PFC population, consistent with the idea that the metacognitive judgement is informed by a readout of mnemonic uncertainty[89]. fMRI evidence obtained in humans also supports the proposal that population-level representations of uncertainty are maintained in the brain areas that track mnemonic content during visual working memory tasks[90].

Other forms of world-directed uncertainty encompass our own bodies. For instance, uncertainty about the extent to which our muscles and joints are able to accomplish particular actions, such as rapidly pointing to a target[91]. Compared to sensory or mnemonic uncertainty, less is known about how motor and action uncertainty is tracked and read out for use in online metacognitive evaluation. However, in tasks requiring skilled action, this is likely to be a key contributor to confidence formation[92,93]. In humans, motor kinematics have been shown to impact upon error-related brain activity[94] and aspects of action production including speech prosody, response time and movement time have all been linked to confidence[95–97].

Another influence of action on metacognitive evaluation concerns memory for one's own decisions and actions. Because the evidence contributing to confidence estimates may be subject to further processing (either parallel or serial, and potentially in downstream brain regions), the decision or action that was actually taken often needs to be stored for later use in self-evaluation[51]. Consistent with this proposal, in human neuroimaging studies, activity in the frontopolar and insula cortex, and enhanced functional connectivity between motor and frontal cortex, was increased when self-action information was available to inform metacognitive judgements[98,99]. One implication of this computational constraint is that metacognition can be improved by making initial choices explicit, a prediction borne out by experiment[98–100].



Taken together, these studies support a computational scheme in which effective metacognition involves tracking world-directed uncertainty in a self-directed format to enable adaptive control of behaviour[7,51,74].

Timescales of metacognition. Metacognitive processes operate dynamically, over multiple interacting timescales: local metacognitive processes track moment-to-moment performance, whereas global metacognition encompasses self-estimates of skill or ability that are maintained

Fig. 3 | Overview of static and dynamic models of metacognition. a, A static signal detection theory-based framework for confidence judgements. Upper panels: according to classical signal detection theory, a decision variable is obtained from either a signal (solid line) or noise (dotted line) distribution and compared with a criterion (*c*) to make a decision. In the example shown, the correct decision is obtained for above-criterion responses (blue-shaded areas). When evidence is weak (that is, when there is more overlap between the signal and noise distributions), the chances of making an error (the red shaded areas) are greater than when the evidence is strong. Confidence reflects the distance of the decision variable from the criterion. Lower panel: when averaging over trials, this produces a characteristic 'folded-X' pattern: as evidence strength increases, confidence decreases for errors and increases for correct decisions. **b**, This folded-X signature has been observed in neuronal firing in the rodent orbitofrontal cortex (OFC) for auditory and olfactory decision tasks[41,53]. Both tasks involved an initial perceptual decision followed by an unpredictable time interval for reward delivery, during which firing in the OFC was measured. The amount of time the animal was willing to wait for a reward before moving on to start the next trial is a proxy of confidence. **c**, Schematic representations of the ways in which dynamic models of decision-making accommodate variability in metacognitive judgements. Left: pre-decisional accounts propose that the dynamics of evidence accumulation before a decision are monitored to inform confidence. Middle: post-decisional accounts propose that confidence is informed by the continued accumulation of evidence after a choice has been made. Right: parallel accounts propose that decisions and confidence estimates are formed by accumulating evidence in parallel neuronal populations. **d**, Empirical evidence in humans and monkeys supports a role for all three dynamic signatures in metacognitive evaluation. Left: electroencephalogram data from humans demonstrating that the slope of the centroparietal positivity (CPP) correlates with perceptual decision confidence before a decision being made, consistent with pre-decisional accounts of confidence formation[66], are shown. Black points below the plot denote the centre of 200-ms time windows in which a linear effect of confidence on CPP slope reached significance ($P < 0.05$) after correcting for multiple comparisons across time. Middle: human electroencephalogram data demonstrating that a second-stage CPP is triggered when errors are made, consistent with post-decisional accounts of confidence formation, are shown. Deeper colours indicate faster error detection responses[81]. The vertical dashed lines represent median response times (RTs). The grey markers indicate timepoints when linear regression of RT on signal amplitude reached significance ($P < 0.05$); black markers indicate centre of 150-ms time windows in which regression of RT on signal slope reached significance ($P < 0.05$; one-tailed predicting steeper slope for faster RTs). Right: prediction accuracy for decoders fit to monkey lateral intraparietal cortex neuronal population data predicting choice (purple) and peri-decision wagers (yellow) as a function of time and aligned to motion onset and saccade onset[64] is shown. The parallel onset and evolution of choice and wager (confidence) decoders is consistent with parallel accounts of confidence formation. The coloured bars at the top indicate when the accuracy for the corresponding decoder was significantly greater than chance (one-tailed Wilcoxon rank-sum test with Šidák's correction). The black bar indicates when prediction accuracy was significantly different for choice versus wager (Wilcoxon signed-rank test with Šidák correction). Part **a** adapted with permission from ref. 52, CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/). Part **b** adapted with permission from ref. 53, Elsevier. The left panel of part **d** reprinted with permission from ref. 66, Sage. The middle panel of part **d** reprinted with permission from ref. 81, CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/). The right panel of part **d** reprinted from ref. 64, Springer Nature Limited.

over days, weeks or even years[101]. There has been increasing interest in understanding the interplay between these timescales of metacognition.

Initial findings in humans documented 'confidence leak' between trials, an autoregressive phenomenon in which confidence on previous trials predicts confidence on subsequent trials, despite the trials being independent[102]. These observations hinted at slower dynamics to metacognition that may be obscured by classical single-trial analyses[102]. Laboratory studies have found that fluctuations in local confidence from trial to trial significantly predict global confidence estimates[103,104], with computational modelling indicating that people learn both from external feedback and local confidence to construct a self-model of their skills and abilities across a range of task domains[105]. Combining a local-global confidence integration task with fMRI revealed a posterior-to-anterior gradient within the PFC: pMFC (dorsal anterior cingulate cortex) activity was sensitive to local (but not global) confidence whereas more anterior ventromedial PFC was modulated by both local confidence and by the level of global belief in performance across blocks[106].

In a Bayesian framework, global confidence may reflect a higher-level prior on the precision of perceptual and cognitive representations[85,107]. Oscillatory activity in the form of neuronal oscillations at alpha and beta frequencies has been linked to priors on confidence in both auditory[108] and visual[109] perceptual tasks in humans. However, compared to 'bottom-up' influences on confidence formation, less is known about the neural basis of these contextual influences on metacognitive judgements[110,111]. One idea is that metacognitive priors are inherited from our sociocultural milieu and possibly only loosely connected to the task at hand[112].

An integrated model of metacognition

We can now tentatively advance a working hypothesis for the neural basis of metacognition in simple cognitive and perceptual tasks (Fig. 4). The sensory and motor cortices contain rich information about task-specific uncertainties, which are tracked and transformed into a self-directed frame of reference in downstream circuits supported by the parietal and prefrontal association cortices. Dynamic models of confidence formation suggest that the brain extracts, in real time, estimates of decision reliability that are informed by the unfolding perception–action cycle — a computation that is linked to EA signals in the parietal cortex. In turn, self-directed (propositional) confidence is represented by multimodal areas of the PFC (insula and medial PFC, together with their cortico-thalamic and cortico-striatal loops) and further broadcast to the anterior PFC for use in self-regulation and public communication. These integrated, cross-domain representations of confidence are likely to be informed by broader cues and priors that (in humans, and possibly other animals) are supported by a 'self-model' that shares neural machinery with social cognition[40,113]. Finally, a posterior-to-anterior gradient within the human PFC may support the more stable, long-run metacognitive estimates that underpin global self-beliefs about skill and ability.

There is evidence of notable commonalities in this architecture across species. First, both human and monkey metacognitive judgements in perceptual tasks are sensitive to fluctuations in uncertainty estimated from neural populations in the early visual cortex[58,86]. Second, in studies of perceptual decision-making, confidence is intimately linked to the properties of neural EA pre- and post-decisionally in both humans and monkeys[37,64–66,74]. Third, it is striking that medial aspects of agranular PFC carry domain-general confidence or error signals in mice, monkeys and humans[53,75,88,114]. Finally, there is evidence for a causal contribution of anterior/lateral aspects of the granular PFC to metacognitive judgements in both humans and monkeys[38,44,47].

There are, however, salient differences between species (Fig. 4b). Findings of PFC involvement in metacognition seem more prevalent

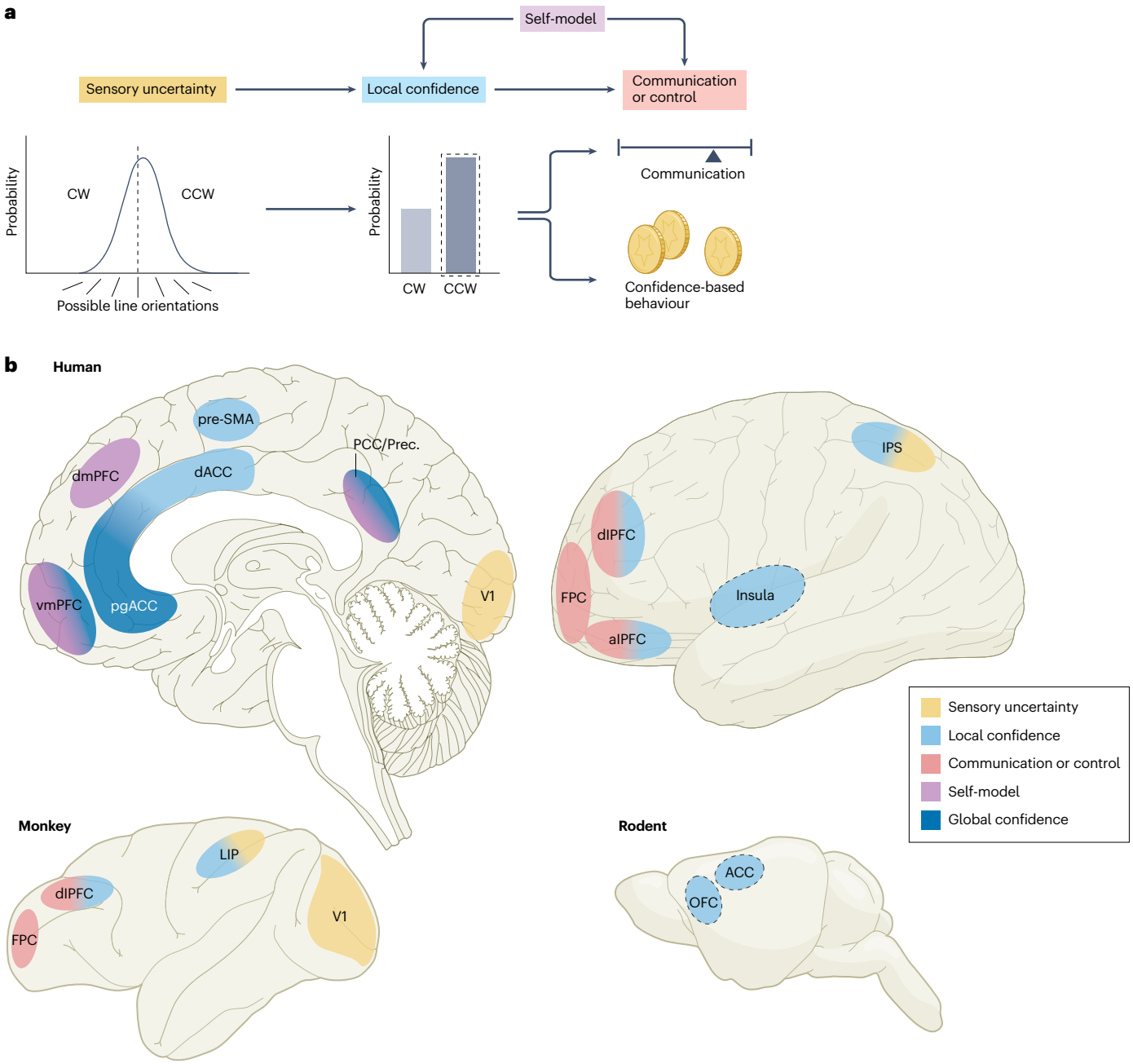


Fig. 4 | Towards an integrative neuroscience of metacognition. a, A graphical illustration of the components of a perceptual metacognitive judgement. A belief about possible world states (here, the curve represents a probability distribution over possible line orientations) generates sensory uncertainty, which is converted into (local) propositional confidence with regard to a categorical decision of whether the stimulus is tilted clockwise (CW) or anticlockwise (CCW). A propositional confidence estimate is then readout for communication or metacognitive control (confidence-based behaviours). Background beliefs about a range of factors influencing self-performance are furnished by a self-model and modulate the formation and usage of confidence. **b**, An illustration showing how the different computational components of metacognition map onto the functional anatomy of humans, monkeys and rodents. Not shown are cortico-basal ganglia and cortico-thalamic loops or macaque medial frontal areas linked to error monitoring. Regions are colour-coded by their association with different computational signatures in the literature so far; where more than one signature has been associated with a particular region, a colour gradient is used. ACC, anterior cingulate cortex; OFC, orbitofrontal cortex; dlPFC, dorsolateral prefrontal cortex; FPC, frontopolar cortex; LIP, lateral intraparietal area; IPS, intraparietal sulcus; alPFC, anterolateral prefrontal cortex; PCC, posterior cingulate cortex; Prec., precuneus; pre-SMA, pre-supplementary motor area; dACC, dorsal anterior cingulate cortex; dmPFC, dorsomedial prefrontal cortex; vmPFC, ventromedial prefrontal cortex; pgACC, perigenual anterior cingulate cortex. Part **a** adapted with permission from ref. 7, CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/).

in human studies than non-human animal studies. There are several potential reasons for this. First, there may be a sampling bias in the types of neural recordings used in animal studies of confidence and metacognition. In support of this, when fMRI was used to obtain whole-brain coverage in non-human primate research, the causal importance of the anterior PFC for metacognition in macaque monkeys was uncovered[38,46]. Second, reliance on post-decisional, explicit reports in human studies may emphasize the recruitment of prefrontal areas sensitive to self-action and conscious self-reflection. This highlights the importance of developing tasks to compare the neural basis of both implicit and explicit metacognitive measures in human studies, to enable more appropriate comparisons to be made with neural signatures obtained in other species.

Finally, it is possible that, even when task features and metacognitive measures are matched, different species may have adopted different algorithmic solutions to metacognitive performance. This possibility was recently supported by a study that compared the neural basis of metacognition across humans and monkeys performing the same prospective confidence task[115]. The task involved comparing an estimate of (future) self-performance in a motion discrimination task against a fluctuating probability of receiving an external reward. Using fMRI, it was shown that the monkey lateral PFC coded self-performance and external reward probability in different subregions. In humans, however, these two latent quantities were jointly tracked in the lateral anterior PFC (Brodmann area 47 (ref. 116)). By comparing the same task across species, insights into the aspects of metacognitive machinery that are conserved can be exposed, potentially providing information that can guide novel interventions to remedy metacognitive dysfunction.

Functional roles of metacognition

Once metacognitive estimates of self-performance are formed, they can be used in the guidance of future behaviour. As noted above, when compared to the increasingly rich picture of metacognitive monitoring that has been provided by neuroscience research, less work has focused on the neural basis of metacognitive control. In this section I explore what is known, and describe opportunities for future work. Like metacognitive monitoring, metacognitive control processes operate over both shorter (local) and longer (global) timescales. Cross-cutting these different timescales are distinct domains of control, which I organize into self and social regulation (Fig. 5).

Self regulation

On a local timescale, metacognition plays an epistemic role in rapid error correction, adaptive regulation of evidence gathering[72,117], selection of the next step in sequential decision-making[118] and the control of information seeking[119–121]. In studies in humans, for example, the current level of confidence during EA has been proposed to regulate how much evidence is required to commit to a decision[72,122]. At slightly longer timescales, fMRI correlates of perceptual confidence in human medial PFC (vmPFC and ACC) were found to predict trial-by-trial variation in curiosity and information seeking[121]. Confidence can also act as a proxy feedback signal when none is available, shaping learning and updating of beliefs[123,124]. In macaque monkeys, for example, dopamine neurons were found to encode teaching signals (reward prediction errors) that scaled in proportion to confidence[125]. It remains to be determined how the machinery of metacognitive monitoring described in earlier sections furnishes the relevant confidence signals for use in learning.

The formation of more global metacognitive estimates is likely to be crucial to allow the strategic arbitration between tasks, or strategies within a task. For instance, if I am deciding whether to invest my time in becoming a musician or a scientist, I might reflect on my past performance in each endeavour to estimate my likely chances of future success. Neurons in monkey pMFC were found to integrate information about previous trials to update long-run performance beliefs and drive decisions about whether to switch strategy[126]. Such arbitration may be supported in part by a 'common currency' for confidence — metacognitive signals that transcend particular tasks, allowing them to be compared in a common frame of reference[53,114,127]. Multiplexed domain-general and domain-specific metacognitive signals could then be readout by downstream regions to support task-level arbitration[128,129] and strategy selection[130]. Evidence for this view comes from studies that have examined the neural correlates of confidence across multiple tasks, finding common currency representations that generalize over task in the medial PFC in both humans[114] and rodents[53]. Other work has implicated the FPC in 'tagging' different first-order behaviours with degrees of confidence or uncertainty to allow efficient arbitration[129,130]. Over more extended timescales, stable beliefs about our skills and abilities may shape our motivation to engage in future endeavours, and impact on feelings of self-worth and self-esteem[101,131].

Social regulation

Metacognitive capacities are central to adaptive social behaviour, allowing both the public sharing of our own confidence and adaptive deferral to others, including artificial intelligence (Box 1). In one study it was found that the sharing of metacognitive estimates between two observers allowed them to perform a psychophysical task together better than either of them could have achieved alone[132]. If we wish to strategically influence group decision-making, it may be advantageous to overstate (or understate) our confidence[133,134]. In a study of the neural basis of these strategic adjustments of 'public' metacognitive statements using fMRI in humans, it was found that, whereas activity in the medial PFC (specifically, pgACC) covaried with private estimates of confidence in a random dot motion decision, the FPC additionally carried information about the extent to which one's confidence level should be adjusted when engaging in public communication[135]. The recruitment of these regions may be sequenced in time: a study using EEG-informed fMRI in humans was able to separate early neural activations correlating with confidence in pgACC from later activations in FPC recruited at the time of metacognitive communication[136].

In turn, a nascent literature has developed to understand how we develop beliefs about the skills and capacities of others: a form of other-directed metacognitive evaluation[7]. Studies in humans have highlighted a role for networks involved in both theory of mind[137,138] and prospective self-simulation (simulating how we would act in similar situations)[139] in this process (Fig. 5b).

Metacognition as a clinical target

Altered metacognition in brain disorders

Dysfunctional metacognition has been hypothesized to underpin symptoms in a range of psychiatric and neurological disorders, and has also increasingly become a target for novel therapeutic interventions.

Disturbances in metacognition have been linked to psychiatric diagnoses of anxiety, depression, obsessive–compulsive disorder and psychosis[140]. For instance, obsessive–compulsive disorder has classically been considered a 'disorder of doubt' in which people lack trust in the veracity of their own perceptual, cognitive and mnemonic processes[141]. By contrast, people with schizophrenia have been characterized as having poor domain-general metacognitive sensitivity[142],

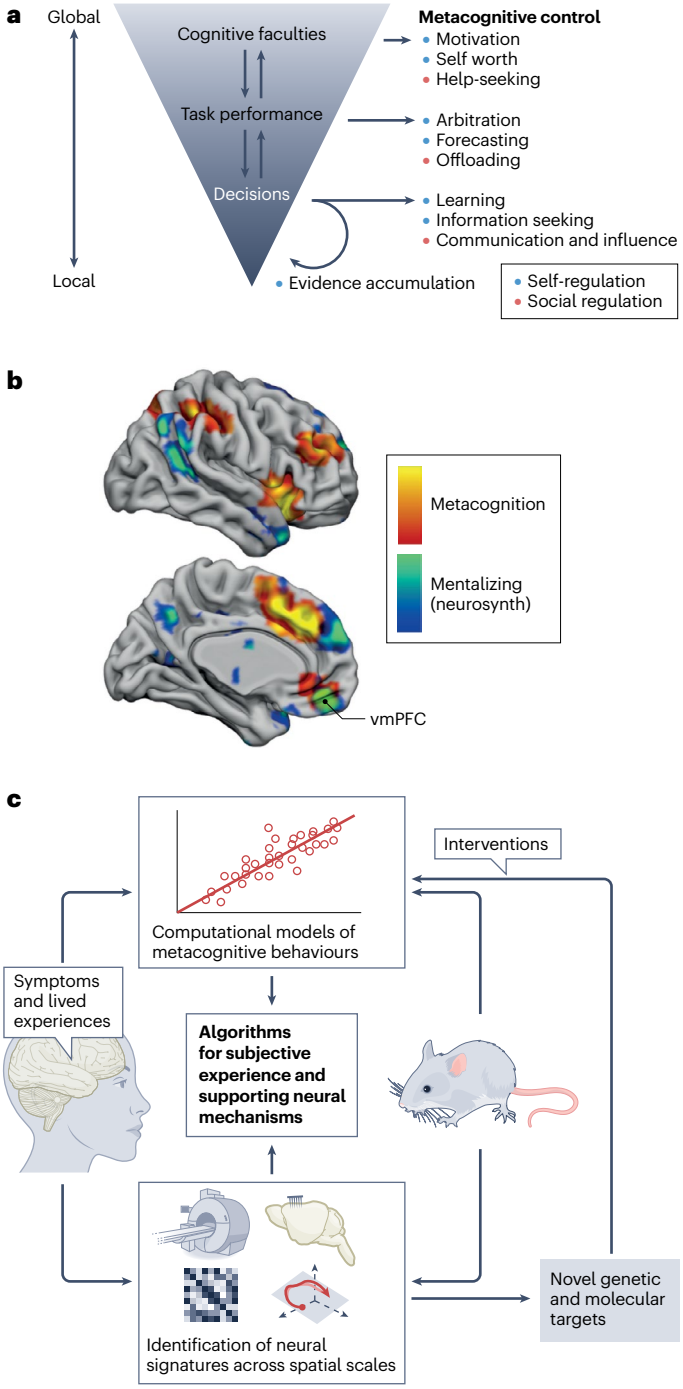


Fig. 5 | New frontiers in metacognition research. a, Extending the hierarchical framework for metacognitive monitoring[101] to incorporate different timescales of metacognitive control. Aspects of control acting on shorter timescales (for instance, on an individual decision) are situated at the local level, whereas aspects acting over longer timescales (for instance, affecting multiple trials) are situated at more global levels. These timescales cross-cut both self-regulatory and social functions of metacognition. **b**, Meta-analyses indicate partial overlap between brain regions engaged in self-directed metacognition and those involved in mentalising (thinking about the mental states of others), particularly within the ventromedial prefrontal cortex[40,184]. The colour scales indicate the strength of associations between activations and metacognition/mentalising. Note that although both maps are corrected for multiple comparisons across the whole-brain volume, the numerical values and thresholds are not comparable, as they are obtained via different meta-analytic methods (activation likelihood estimation for metacognition, cluster-level family-wise error rate corrected at $P < 0.05$; multilevel kernel density analysis for mentalising, false discovery rate corrected at $P < 0.01$). Overlap was computed from voxels that surpassed these thresholds in both maps. **c**, Integrative approaches in metacognitive neuroscience show promise for tackling disorders of subjective experience. Metacognitive measures are consciousness selective and track distortions in subjective reality[185] (Box 2). These measures can be used in conjunction with computational models to identify algorithms (for instance, for confidence formation or reality monitoring) that can be probed in animal models using non-verbal assays. Neural mechanisms supporting these algorithms can be validated across species by harnessing analysis tools such as representational similarity analysis to probe the structure of neural representations across techniques and spatial scale[186]. Knowledge of the implementation of these algorithms in animal models guides novel insights into their genetic and molecular basis, potentially unlocking new interventions to tackle subjective symptoms in humans. Interventions target neurobiological mechanisms to alter parameters of computational models of metacognition. vmPFC, ventromedial PFC. Part **b** adapted with permission from ref. 40, CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/).

and often manifest impaired clinical and cognitive insight into the abnormal nature of their experiences[143].

More recently, transdiagnostic methods have been applied in large general population samples to relate metacognitive profiles to variation in symptom dimensions[144]. One way of organizing these findings is to posit a phenotypic space spanning changes in metacognitive bias and sensitivity across either broad or narrow task domains. For instance, anxiety-depression scores have been associated with lower local confidence (a change in metacognitive bias but not sensitivity[145]), whereas a symptom dimension tracking compulsive behaviours and intrusive thoughts is associated with both overconfidence at the local, trial level, and underconfidence (doubt) at more global levels[146,147] — findings which generalize across distinct task domains[148,149]. When studying links between metacognition and mental health, care is needed to ensure that conclusions are not unduly affected by idiosyncratic biases affecting self-report metrics of both confidence and symptoms[150,151].

A common feature of a range of neurological disorders is lack of awareness of cognitive problems, known as anosognosia[152] and often associated with impairments in prefrontal function. For example, around 75% of people with frontotemporal dementia show impaired awareness of illness[153]. Disparate forms of anosognosia can be considered failures to update a global metacognitive model of personal-level beliefs about one's own abilities[152,154]. Objective measures of metacognitive sensitivity have been linked to levels of tauopathy (a major proteinopathy in Alzheimer disease) in older adults[155], and to changes in prefrontal function in people with substance-use disorder[156]. These impairments in metacognition have functional consequences, such as a reluctance to seek help, take medication, or take steps to compensate for cognitive changes[157]. However, despite its clinical importance, metacognitive capacity is not yet captured by standard neuropsychological assessments. Functional disorders — cases in which people report subjective symptoms in the absence of deficit — perhaps reflect a pure expression of a change in metacognition, without any underlying measurable alteration in cognition or performance. However, few studies have used objective measures of metacognition in functional disorders[158].

An important open question concerns the causal relationships between metacognitive disturbances and clinical symptoms. If metacognition is a root cause of mental ill health, then improving or modifying it may become a viable treatment target. Alternatively, if it is primarily an accompanying symptom, it may instead serve as a novel marker of underlying dysfunction. These questions could be addressed initially with longitudinal data and then with intervention studies.

Interventions

The prevalence of distortions in metacognition in clinical conditions, and the importance of effective metacognition in educational and social settings, has prompted growing interest in the potential for interventions.

Initial work explored whether behavioural training may be effective for increasing metacognitive efficiency, with some evidence that people's confidence reports could be shaped via feedback, although results were variable across studies[159–162]. Neurocomputational modelling is important to understand the locus of action of these effects. In one recent study, the fitted value of meta-uncertainty within the CASANDRE model was lower for (well-trained) monkeys than for human participants on their first visit to the lab[58]. However, humans were able to approach similar levels of metacognitive competence after further training with an additional 2,200 trials of the task. In clinically-focused studies, systematic protocols were developed for metacognitive training that seeks to provide a structured education about common cognitive biases and 'plant the seeds of doubt' in people who are prone to overconfidence and/or lack insight[162]. Metacognitive training has been shown to have consistent effects in reducing delusions and positive symptoms of psychosis, leading to it being recommended in national treatment guidelines in Australia, New Zealand and Germany. How such training may be altering the neural and computational components of confidence formation described above remains to be established.

Other work has shown potential benefits of meditation[163], neurofeedback[164] and brain stimulation for modulating metacognition. Multi-region transcranial magnetic stimulation (TMS) has proven particularly illuminating in probing the causal contribution of particular brain areas. For instance, one study showed that although dlPFC stimulation reduced metacognitive bias (confidence level), FPC stimulation improved metacognitive efficiency[47] (see also refs. 165,166). Another study revealed a double dissociation between the effects of TMS on decision-making and metacognition: neurostimulation aimed at boosting back projections from cortical regions V5/MT to V1/V2 enhanced motion sensitivity without impacting metacognition, whereas boosting intraparietal sulcus-to-V1/V2 back projections increased metacognitive efficiency without impacting motion sensitivity[167]. In a meta-analysis of 25 TMS and transcranial electrical stimulation studies of metacognition in humans, the most robust finding was a contribution of anterior and lateral portions of PFC to metacognition in both perceptual decision-making and memory tasks[168]. As for much exploratory work in the cognitive neurosciences, what is now needed is the combination of these promising methods with computational models that can develop precise hypotheses about the processes being targeted by particular types of stimulation.

Finally, a small number of studies have explored pharmacological effects on metacognition and confidence. Blocking noradrenaline function improved metacognitive efficiency in a perceptual task[169], whereas enhancing catecholaminergic neuromodulation changed perceptual sensitivity, but not metacognition[170]. There is also nascent work on the role of bodily states in multimodal metacognitive judgements, with previous findings indicating that unexpected physiological arousal compromises the quality of metacognitive judgements[171] and suggesting that interoceptive uncertainty informs metacognitive evaluations

Glossary

Adaptive control
The ability to adjust thoughts, decisions or actions as circumstances change. Facilitated by metacognitive monitoring of current performance levels, which allows behaviour to be modified to pursue one's goals.

Bayesian framework
An approach to perception, cognition and metacognition in which the brain is treated as representing beliefs as probability distributions, and updating them by combining prior expectations with incoming evidence weighted by its reliability. Within this framework confidence is expressed as a probability, and expectations about reliability as priors.

BOLD signal
The blood-oxygen-level-dependent signal measured by functional magnetic resonance imaging. It reflects changes in blood oxygenation and flow associated with nearby neural activity, providing an indirect measure of brain activity.

Machine learning
A set of computational methods that identify patterns in data and use them to make predictions or classifications. In neuroscience they are typically used as decoders: a classifier is trained to readout a variable — such as a stimulus feature or its associated uncertainty — from patterns of brain activity.

Metacognitive bias
A difference in subjective confidence despite objective task performance remaining constant.

Metacognitive efficiency
The level of metacognitive sensitivity corrected for differences in first-order performance.

Metacognitive sensitivity
How effectively confidence distinguishes between correct and incorrect judgements. Estimates are often confounded with changes in first-order performance and metacognitive bias.

Meta-uncertainty
Imprecision in an observer's own estimate of the reliability of their perceptual or cognitive system. Because this second-order estimate is itself noisy, it imposes a ceiling on metacognitive performance even when first-order sensitivity is intact. A central parameter of the CASANDRE model.

Neurofeedback
A training technique in which brain activity is measured in real time and displayed back to the participant, who is rewarded for producing a target activity pattern. It offers a route to directly modulating specific neural signals, including those associated with confidence.

Propositional confidence
Confidence in one's own (possibly hypothetical) decisions or actions, expressed as covert propositions such as "I think I will remember this word." Distinguished from uncertainty about states of the world.

Psychometric functions
Curves describing how a behavioural measure changes with the physical strength of a stimulus. Fitting them separately for choices and for confidence ratings can dissociate first-order sensitivity from metacognitive parameters.

Theory of mind
The capacity to attribute mental states such as beliefs, intentions and knowledge to other agents, and to use those attributions to predict or explain their behaviour (also known as mentalising). Partially overlapping brain networks support theory of mind and self-directed metacognition.

across domains[131]. New experimental paradigms now allow for the direct investigation and modelling of metacognition about interoceptive states such as respiration and heart rate[172,173]. Given the potential clinical importance of metacognitive interventions, this is an area ripe for further investigation.

Future directions

Innovating new measures, metrics and models

Advances in measurement have created new opportunities for charting metacognition across populations and task domains. However, the development of these measures and computational models has not been matched by equivalent investment in psychometric evaluations of associated model parameters[174]. In addition, popular frameworks, such as SDT-based and EA-based models, have led to a focus on retrospective confidence estimates in two-alternative forced-choice tasks, precluding the evolution of more naturalistic measures[175,176]. One approach to rectifying this will be to develop more flexible ways of relating task performance to metacognition, such as relative psychometric function analysis[177], or computational approaches that cast metacognition as a problem of optimizing mutual information between behaviour and reflective judgements[178].

The idea that decision confidence should be framed as a probability of a first-order behaviour being objectively accurate or correct has been challenged on both empirical and theoretical grounds. Theoretically, it is often hard to define accuracy for subjective decisions, such as value-based choices or aesthetic judgements, and yet empirical studies show that people hold a meaningful sense of confidence in these choices[29]. Instead, more recent frameworks have suggested that the ground truth for metacognition should not be objective correctness but self-consistency — the probability of making the same choice across multiple presentations of the same decision problem[56,179,180]. Key to the self-consistency approach is that metacognitive estimates arise from an overall evaluation of the quality of the perception–action cycle, rather than a direct evaluation of the sensory input[8].

There is also the question of meta-metacognition: behavioural evidence shows that humans can engage in third or even fourth-order self-evaluations, tracking the reliability of their own confidence judgements[181]. It would be of interest to probe whether the same confidence formation motifs documented here could be applied to metacognitive judgements themselves, providing a powerful engine for recursive self-modelling[182].

Integrative approaches

Metacognitive neuroscience crosses species (from rodents to humans) and levels of analysis (from single neurons to behaviour), while retaining a strong connection to human subjective experience (Box 2). This blend positions metacognition research as potentially transformative for the discovery of conserved circuit-level motifs that underpin disturbances in conscious experience, such as those generating a sense of unreality in psychosis, or persistent underconfidence in anxiety disorders. However, achieving this goal will require greater integration between human and animal laboratories to develop cross-species behavioural assays and computational models to bridge the gap between neural circuits and experience (Fig. 5c). It also remains unknown which aspects of metacognition are shared across species. Rodents lack the granular lateral part of the PFC that is thought to be essential for some metacognitive functions in primates, particularly perhaps those supporting flexible task arbitration and control. However, both rodents and primates possess agranular frontal cortex and insula, which may be important for implicit aspects of confidence and reality monitoring. Fully addressing this question requires consideration of how to integrate neural data across species, in which identifying homologues between circuits and brain regions is fraught.

A goal for future research is to identify common building blocks of metacognition, while being alive to the potential that certain aspects, such as explicit self-knowledge, may be uniquely human. It will also be of interest to expand the study of the neural basis of metacognition beyond rodents and primates in order to probe minimal circuits that may support forms of metacognitive computation — for instance, the capacity to opt-out of difficult choices displayed by honeybees[183]. This endeavour will benefit from adopting performance-controlled assays of metacognition that have been developed in work on mammals. Even if some elements of metacognition may be restricted to primates or even humans, the tools provided by metacognition research offer a bridge from circuit-level analysis in animal models to subjective symptoms in humans and harbour considerable potential for an integrative neuroscience of psychopathology.

Conclusion

Metacognition is no longer restricted to philosophical or psychological enquiry. Across species, convergent behavioural, computational and neural evidence indicates that self-evaluation arises from structured transformations of uncertainty into a format that can be used for adaptive control. Dynamic EA processes provide candidate building blocks for local confidence, whereas multimodal prefrontal circuits integrate these signals into more abstract, global self-beliefs. This architecture supports adaptive control at multiple timescales, from rapid error correction to strategic task arbitration and social communication.

Important open questions remain. We lack precise psychometric characterization of many metacognitive metrics, and cross-species tasks are still only partially aligned. The field must also clarify which aspects of metacognition are evolutionarily conserved and which depend on uniquely human forms of self-modelling. Nonetheless, the integrative framework emerging from recent work suggests that metacognition research offers a tractable computational handle on subjective experience. By linking circuit-level mechanisms to distortions in confidence and self-belief, metacognitive neuroscience offers a principled route towards understanding — and potentially remediating — disorders of subjective reality.

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